

Akshay Shankar

COMPUTATIONAL PHYSICIST

Department of Physics and Astronomy, Ghent University, Gent, Belgium

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Research Focus: Developing numerical algorithms to model ultracold atoms and realize their potential for practical quantum applications.

Summary

PhD Student at Ghent University specializing in Tensor Network simulations for one-dimensional quantum gases in the continuum. Experienced in developing high-performance numerical algorithms (Julia/Python) to study quantum many-body systems. Also experienced in building interactive visualizations and user interfaces using front-end web development tools and game engines.

Skills

Scientific Programming	Julia (TensorKit.jl, MPSKit.jl), Python (NumPy, SciPy), C++, Git, Linux
Numerical Methods	Tensor Networks (DMRG, VUMPS, TDVP), Quantum Monte Carlo (SSE), Mean-field theory (Gutzwiller, Cluster-MFT)
Visualization & UI	Godot Engine (GDScript), Blender, Inkscape, Front-end (HTML, CSS, JavaScript)

Research Experience

Ph.D. Project: Finite-element matrix product states for continuum models

Ghent University

SUPERVISED BY PROF. KAREL VAN ACOLEYEN AND PROF. JUTHO HAEGEMAN | JULIA

Sept 2023 - Present

- Developed a theoretical framework to utilize Matrix Product States (MPS) with *non-orthogonal* single-particle orbitals to study non-relativistic quantum field theories in one dimension.
- Applied the framework to capture ground state physics of the Lieb-Liniger model using a finite-element basis and a multi-grid optimization scheme to efficiently reach the continuum limit.
- Developed a performant, open-source numerical implementation of the Finite-element Matrix Product States (FE-MPS) formalism involving custom DMRG code built on top of MPSKit.jl.

M.S. Thesis: Low-temperature phases of interacting bosons in a lattice

IISER Mohali & University of Stuttgart

SUPERVISED BY DR. SANJEEV KUMAR AND PROF. TILMAN PFAU | JULIA, PYTHON

June 2022 - Apr. 2023

- Implemented various many-body algorithms (Cluster Mean-Field Theory, Stochastic Series Expansion, Neural Quantum States) to map the phase diagram of the extended Bose-Hubbard model on a square lattice.
- Collaborated with experimentalists at the University of Stuttgart to guide the exploration of supersolid phases arising from Dysprosium atoms with dipole-dipole interactions in an optical lattice.

Research Internship: Simulating BEC dynamics in one dimension

IIT Kharagpur (Remote)

SUPERVISED BY DR. VISHWANATH SHUKLA | JULIA, FORTRAN90

May 2021 - Aug. 2021

- Developed a 1D Gross-Pitaevskii Equation (GPE) solver to simulate ground-state physics and real-time dynamics of Bose-Einstein Condensates.
- Investigated parallel computing frameworks (MPI, OpenMP) to optimize the implementation.

Education

Ghent University

Gent, Belgium

PH.D. IN PHYSICS

Oct. 2023 - Present

- Supervised by Prof. Karel Van Acoleyen and Prof. Jutho Haegeman.

Indian Institute of Science Education and Research (IISER), Mohali

Punjab, India

B.S.-M.S. INTEGRATED DEGREE (MAJOR IN PHYSICS)

Aug. 2018 - June. 2023

- Cumulative Performance Index (CPI): 10.0/10.0**

Honors & Awards

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| 2023 | President's Gold Medal , for the best academic performance in the class of 2023. | IISER Mohali |
| 2019 | CNR Rao Foundation Award , for obtaining 10.0 SPI in the 2nd semester. | IISER Mohali |
| 2018 | Innovation in Science Pursuit for Inspired Research - Scholarship for Higher Education (INSPIRE-SHE) , for top 1% performance in AISSCE. | DST India |

Publications & Preprints

Finite-Element Matrix Product States for Continuum Models in One Dimension	Arxiv
AKSHAY SHANKAR, KAREL VAN ACOLEYEN, AND JUTHO HAEGEMAN	Jun. 2026
https://arxiv.org/abs/2606.14873	

Open Source Software

TentMPS.jl	
IMPLEMENTATION OF THE FE-MPS FORMALISM DEVELOPED IN ARXIV:2606.14783.	2026
LiebLinigerBetheAnsatz.jl	
IMPLEMENTATION OF THE BETHE ANSATZ SOLUTION FOR THE LIEB-LINIGER MODEL.	2025
ThePhysicsHub	
REPOSITORY OF EDUCATIONAL INTERACTIVE SIMULATIONS DEVELOPED BY A GLOBAL COMMUNITY OF STUDENTS.	2022

Conferences & Workshops

2026	Challenges of Low-Dimensional Quantum Gases: Brazilian-German Wilhelm and Else Heraeus Seminar , presented a poster titled "Finite element matrix product states for continuous systems"	ICTP-SAIFR, Sao Paulo, Brazil
2024	Les Houches Predoc school on cold atoms , presented a poster titled "RadialMPS: A novel scheme to study 2D Bose gases in the continuum"	Les Houches, France
2024	European tensor network school , presented a poster titled "RadialMPS: A novel scheme to study 2D Bose gases in the continuum"	Padua, Italy